
Autonomous Production Choke Controller

Execution Guide

Honeywell Hackathon 2026

Engineering Submission

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Quick Start (2 minutes)

```
cd 01_Source_Code
pip install -r requirements.txt
python run_scenarios.py
streamlit run dashboard.py
```

Step-by-Step Instructions

Step 1: Install Dependencies

```
pip install -r requirements.txt
```

Required: numpy>=1.24.0, matplotlib>=3.7.0, streamlit>=1.28.0, plotly>=5.17.0

Step 2: Run the Control Pipeline

```
python run_scenarios.py
```

Executes: (1) System Identification - 6 automated step tests, (2) Model Identification - FOPDT 63.2% method, (3) Control Scenarios - 3x50 steps, (4) Plot Generation - 4 PNGs in output/.

Expected terminal output shows PHASE 1/2/3 sections and a PERFORMANCE SUMMARY with tracking errors and constraint violation counts.

Step 3: Launch the Dashboard

```
streamlit run dashboard.py
```

Opens <http://localhost:8501> with 8 interactive pages: Home, Architecture, Scenario Playback, Performance KPIs, Constraint Monitor, Model Summary, Validation Report, Export Results.

Step 4: Generate All Deliverables

```
python generate_artifacts.py
```

Produces publication-quality figures: 9 step test plots, 2 architecture diagrams, 3 enhanced scenario plots, 2 model validation plots.

Configuration

Table 1: Key configuration parameters.

Parameter	Default	Description
ts	1.0 h	Control interval
ramp_limit	5.0%	Max choke movement
prediction_horizon	3	Steps to predict
candidate_step	1.0%	Coarse resolution
fine_step	0.5%	Fine near target
weight_effort	0.01	Effort penalty
weight_margin	0.005	Margin reward
safety_margin	0.05	5% constraint buffer
bias_alpha	0.3	EWMA bias factor

Constraint limits: WHP[200,600], FLP[150,500], BHP[2500,3500] psi, set in run_scenarios.py.

Project Structure

01_Source_Code/ contains 17 Python modules: __init__.py (public API), config.py, models.py, controller.py (orchestrator), process_monitor.py, target_manager.py, predictor.py, safety_gate.py, selector.py, simulator_adapter.py (TestSimulator), step_test.py, model_identifier.py, plotter.py, run_scenarios.py (entry point), generate_artifacts.py, dashboard.py (Streamlit app), and requirements.txt.

Expected Validation Results

Table 2: Expected results at seed=42.

Scenario	Final Q	Target	Error	Violations
A: Constant Target	~81	80	~1	0
B: Target Change	~121	120	~1	0
C: Infeasible	~186	250	~64	1 (boundary)

Troubleshooting

Table 3: Common issues and solutions.

Issue	Solution
Import error	Run: pip install -r requirements.txt
Dashboard won't launch	Check streamlit is installed
Plots not generating	Ensure output/ directory exists
Different results	Verify seed=42 in TestSimulator